Prevalence and Control of Diabetes Among US Adults, 2013 to 2023

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February 27, 2025

Table. Prevalence of Diabetes Among US Adults, 2013-2023

	Weighted prevalence, % (95% CI)[a]			
	2013-2014 (n = 5347)	2015-2016 (n = 5155)	2017-2020 (n = 8022)	2021-2023 (n = 5739)
Overall prevalence	12.8 (11.8-13.9)	14.3 (12.6-16.1)	14.5 (13.5-15.5)	14.1 (12.7-15.4)
Age group, y				
20-44 (n = 9271)	4.2 (3.4-5.0)	5.8 (4.6-7.0)	5.1 (4.3-5.8)	5.0 (3.9-6.0)
45-64 (n = 8561)	17.3 (14.6-19.9)	18.9 (15.9-21.8)	19.1 (17.1-21.1)	19.3 (16.9-21.7)
≥65 (n = 6431)	26.1 (23.7-28.5)	27.0 (23.6-30.4)	29.7 (27.2-32.1)	26.9 (23.0-30.9)
Sex				
Male (n = 11 557)	13.8 (12.3-15.3)	16.1 (13.7-18.6)	15.9 (14.3-17.5)	15.1 (13.3-16.9)
Female (n = 12 706)	11.9 (10.8-13.0)	12.6 (11.0-14.2)	13.3 (11.7-14.9)	13.1 (11.2-15.0)

[a] Nationally representative estimates of outcomes after weighting. Diabetes was defined as a hemoglobin A_{1c} level 6.5% or greater, fasting plasma glucose level 126 mg/dL (6.99 mmol/L) or greater, or a health care diagnosis of diabetes.

Although diagnosis and management of chronic conditions were affected by disruptions in care access and health behaviors during the COVID-19 pandemic, little is known about how the burden of diabetes changed in recent years.[1] This study evaluated trends in the prevalence and control of diabetes among US adults overall and by age and sex between 2013 and 2023.

Methods:

We analyzed data from the National Health and Nutrition Examination Survey (NHANES) between 2013 and August 2023. NHANES is a nationally representative survey of noninstitutionalized US civilians conducted through in-home interviews and mobile examination centers. Response rates decreased from 68.5% in 2013-2014 to 25.7% in 2021-2023, and survey weights adjusted for nonresponse. We included nonpregnant adults 20 years or older with available hemoglobin A_{1c} (HbA_{1c}) measurements. Information about age and sex was collected through established questionnaires. Diabetes was defined as an HbA_{1c} level 6.5% or greater, fasting plasma glucose level 126 mg/dL (6.99 mmol/L) or greater, or a health care diagnosis of diabetes. Glycemic control was defined as an HbA_{1c} level less than 7.0%.

First, we estimated the prevalence of diabetes for each survey cycle across the study period. Prevalence estimates were standardized by age and sex using the direct method to the 2010 US Census population. Second, we evaluated mean HbA_{1c} levels and glycemic control rates among individuals with a health care diagnosis of diabetes overall and across age and sex subgroups. Changes across survey cycles were estimated using multivariable linear regression with indicator variables for the survey cycle (ie, 2013-2014, 2015-2016, 2017-2020, and 2021-2023), age, sex, and race and ethnicity.

All analyses incorporated NHANES survey weights to obtain nationally representative estimates. Analyses were conducted using R version 4.2.3 (R Foundation). Statistical significance was defined as 2-sided $P < .05$. The NHANES protocols were approved by the National Center for Health Statistics, and participants provided written informed consent.

Results:

The unweighted study population included 24 263 adults (weighted mean age, 48.5 years; 51.4% female), with 4623 identified as having diabetes. The prevalence of diabetes did not significantly change, from 12.8% (95% CI, 11.8%-13.9%) in 2013-2014 to 14.1% (95% CI, 12.7%-15.4%) in 2021-2023 ($P = .25$) (Table).

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Among adults with a diagnosis of diabetes (3541/4623 [77%]), while mean HbA_{1c} levels were stable between 2013-2014 and 2017-2020 (P = .37), they significantly increased from 7.31% (95% CI, 7.15%-7.48%) in 2017-2020 to 7.60% (95% CI, 7.44%-7.76%) in 2021-2023 (P = .03). Similarly, glycemic control rates were stable between 2013-2014 and 2017-2020 (P = .12*)* but declined from 54.3% (95% CI, 46.9%-61.7%) in 2017-2020 to 43.5% (95% CI, 38.5%-48.5%) in 2021-2023 (P = .02).

Increases in mean HbA_{1c} levels were concentrated among young adults aged 20 to 44 years (7.43% [95% CI, 6.94%-7.92%] in 2017-2020 to 8.51% [95% CI, 7.86%-9.16%] in 2021-2023; P = .03) (Figure). Young adults also experienced a significant decline in glycemic control rates (57.4% [95% CI, 46.1%-68.7%] in 2017-2020 vs 37.1% [95% CI, 26.9%-47.3%] in 2021-2023; P = .006). Similar patterns were not observed across other age or sex subgroups.

Figure. Mean HbA_{1c} Level and Glycemic Control Among Adults With a Diagnosis of Diabetes by Age and Sex, 2013-2023

A and C, Mean hemoglobin A_{1c} (HbA_{1c}) levels among subgroups of adults with a health care diagnosis of diabetes (unweighted n = 3541) from the National Health and Nutrition Examination Survey (NHANES), 2013-2014 to 2021-2023. B and D, Nationally representative estimates of the prevalence of glycemic control, defined as an HbA_{1c} level less than 7.0% among subgroups of adults with a health care diagnosis of diabetes. Unweighted sample sizes were 351 for adults aged 20 to 44 years, 1535 for those aged 45 to 64 years, 1655 for those 65 years or older, 1848 for males, and 1693 for females. Error bars indicate 95% confidence intervals.

Discussion:

This study found that the prevalence of adults with diabetes did not significantly change between 2013 and 2023, but glycemic control among those with diagnosed disease worsened in 2021-2023 after nearly a decade of stability. This trend was most pronounced among young adults. The increase of 1% in mean HbA_{1c} levels and 20% decrease in glycemic control would increase the lifetime risk of cardiovascular events.[2]

Potential explanations for these findings include increased sedentary behavior, reduced social support, heightened mental health stressors, and limited access to health care and medications during the pandemic.[1] A prior national study found that younger adults with diabetes were significantly more likely to report disruptions in health care access and difficulty accessing diabetes medications than their older counterparts during the pandemic.[3] In contrast, older adults have near-universal insurance coverage under Medicare, which may have helped maintain access to care and medications.

Study limitations include the low NHANES response rate during the 2021-2023 survey cycle, although weighting adjustment methods applied to this cycle addressed potential nonresponse bias.[4] Race and ethnicity were not analyzed due to changes in sampling, the number of young adults was small, and no information on medication use was available.

Given the recent stagnation in mortality improvement among young adults with diabetes[5] and the increased risk of adverse cardiovascular outcomes associated with poor glycemic control,[6] public health and policy efforts are needed to improve diabetes control, especially in younger populations.

Article Information

Accepted for Publication: December 20, 2024.

Published Online: February 27, 2025. doi:10.1001/jama.2024.28513

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Drafting of the manuscript: Inoue, Liu.

Critical review of the manuscript for important intellectual content: All authors.

Statistical analysis: Inoue, Liu, Wadhera.

Administrative, technical, or material support: Liu, Aggarwal.

Supervision: Wadhera.

Conflict of Interest Disclosures: Dr Inoue receives research support from the Japan Society for the Promotion of Science (23KK0240), the Japan Science and Technology (JST PRESTO; JPMJPR23R2), the Japan Agency for Medical Research and Development (AMED; JP22rea522107), and the Program for the Development of Next-generation Leading Scientists with Global Insight (L-INSIGHT) sponsored by the Ministry of Education, Culture, Sports, Science and Technology (MEXT), Japan. Dr Wadhera receives research support from the

National Heart, Lung, and Blood Institute (R01HL164561, R01HL174549) and the National Institute of Nursing Research (R01NR021686) at the National Institutes of Health, American Heart Association Established Investigator Award, and the Donaghue Foundation. He also serves as a consultant for Abbott, CVS Health, and Chambercardio outside the submitted work. No other disclosures were reported.

Funding/Support: This work was supported by the National Heart, Lung, and Blood Institute (R01HL164561) and the American Heart Association Established Investigator Award (Grant #24EIA1258487).

Role of the Funder/Sponsor: The National Heart, Lung, and Blood Institute and American Heart Association had no role in the design and conduct of the study; collection, management, analysis, and interpretation of the data; preparation, review, or approval of the manuscript; and decision to submit the manuscript for publication.

Data Sharing Statement: See the Supplement.

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